Sahaj Bhadja

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Summary

Software Engineer with 2+ years building data pipelines and concurrent backend systems in **Python** and **C++** on **Linux**, with time-series data processing, automated testing, performance tuning, and simulation-based machine learning research.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Advanced NLP, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Object-Oriented Programming, Software Engineering, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Built a fault-tolerant **concurrent** job pipeline in Python with background workers, explicit state transitions, and failure isolation, processing 1K+ jobs at 3x throughput without cascading failures under concurrent production load.
- Built ingestion pipelines with incremental loads, **data validation**, and reconciliation checks across PostgreSQL, SQL Server, and cloud storage, sustaining 99%+ reliability over 5K+ monthly production jobs for 500+ users.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Designed a **multithreaded** batch-processing engine in Python with a thread pool and fault-tolerant execution, processing 1K+ files per run while keeping APIs stable and responsive under peak production load.
- Built a production backend on AWS handling 10K+ daily requests for 150+ instructors, **debugging** OCR timeouts and model failures with fallback routing and cutting feedback turnaround 60%.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Built Python pipelines processing 100K+ **time-series** trajectory logs sampled at 10Hz, aligning multi-source signals to a global coordinate frame for training and evaluation of a PyTorch planner.
- Designed a spatial-temporal attention encoder with distance-aware interaction weighting over map geometry and agent tracks, benchmarking it against three baselines in **closed-loop simulation** and improving planning stability 15%.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Automated nightly builds that compiled the latest code, installed executables on **Linux**, Windows, and macOS test machines, and ran 20+ unit and integration suites with failure reports.
- Built object-oriented modules with Singleton and Factory patterns and **optimized performance** of API integrations with caching and pagination, improving response times 30% and cutting technical debt 20%.

Cyber Security and Data Mining: Detecting Malicious URL **May 2023 – August 2023**
Data Analyst – Research Fellowship *India*

- Engineered 21 behavioral **features** over a 650K+ URL dataset and trained Gaussian Naive Bayes and Random Forest classifiers to detect phishing and malware from statistical signal patterns.
- Compared classical models against RNN, LSTM, Bi-LSTM, GRU, and CNN architectures, achieving **96%+ detection accuracy** and documenting trends in phishing and malware behavior across the full dataset of URLs.

Skills

Languages: C++, Python, Java, SQL, Bash, Shell, JavaScript
Systems: Linux, Concurrency, Multithreading, Operating Systems, Data Structures, Algorithms, Performance Optimization
Data & ML: Data Pipelines, Time-Series Processing, Data Validation, PyTorch, TensorFlow, Feature Engineering, Simulation
Testing & Tools: Unit Testing, Integration Testing, PyTest, Debugging, Git, CI/CD, Jenkins, Docker, Postman
Storage & Services: PostgreSQL, SQL Server, MySQL, FastAPI, Flask, REST APIs, AWS, TCP/IP, MongoDB, GitHub Actions

Projects

Multi-Sensor Acquisition Pipeline | *C++17, CMake, gRPC, Protocol Buffers, GoogleTest, Linux*

- Designed a **C++17** pipeline that reads several serial and Ethernet sensors on separate threads, timestamps samples into lock-free ring buffers, and streams processed frames to clients through **gRPC**.

Sensor Data Validation Toolkit | *Python, NumPy, pandas, PyTest, perf, Valgrind*

- Designed **Python** scripts that replay recorded sensor runs, flag dropped samples and drift, compare results against reference measurements, and profile the C++ pipeline with **perf and Valgrind**.

CPU-Scheduling Simulator Live | *C++, React.js, JavaScript, EJ-2, OS Scheduling*

- Implemented FCFS, SJF, SRT, Round Robin, priority, and multilevel queue **scheduling** in C++ with Gantt visualization, testing 1K+ process combinations and cutting average turnaround time 25% across test scenarios.